

SWITCHBACK DESCENT

A P R E P R I N T

AUXI & CAITLYN MEEKS

AuxiLab Tenerife

research@auxi.cafe

August 11, 2026

registered 2026-08-11

run ledger #10–#11

2 seeds × 2 terrains, 18 fits each

replicated before publication

promoted 2026-08-11

ABSTRACT

The follow-up brief to Goat Descent was essentially one word: *switchbacks*. We built the trail literally — a unit step tilted downhill at a fixed grade, traversing across the slope and hairpinning every L steps — and asked whether traverse amplitude is a second safety axis, independent of the learning rate. Three results, each holding its registered inequalities on both seeds — though the registered headline prediction itself missed, informatively. (1) At matched effective downhill stride $\eta_{\text{eff}} = s \cdot \text{grade}$, small-amplitude switchbacks are **indistinguishable from straight descent**: all 16 small-amplitude cells (2 seeds × 8) tie within ± 0.01 bits — the grade cap prices as a learning rate and nothing else. (2) The registered traverse-tax prediction missed, informatively: the tax attaches to **slow** legs, not wide ones. Hairpinning every 64 steps costs $+0.076$ – 0.090 bits on both terrains, while stride-built amplitudes as large as the taxed cells' are free. (3) On the ill-conditioned terrain, wide *fast* traverses **beat** their matched straight descent by 0.010 – 0.025 bits in all four cells — orthogonal dither rescues part of the big-stride thrash penalty, at exactly the operating point where the original goat study caught the lab's divergence tripwire sleeping.

Keywords stochastic optimization · normalized gradient methods · momentum · learning rate · gradient noise · pre-registration

§1 The question

A mountain trail never walks the fall line down a steep face; it descends at a capped grade, traversing across the slope and reversing in hairpins. Goat Descent established what fixed-stride, normalized steps buy on the lab's two feature terrains; its notes named, but did not build, the capped-grade trail. This study builds it, to answer a question the metaphor poses precisely: at *matched effective downhill rate*, does zigzagging cost anything? Is traverse amplitude a second safety axis, as on a real mountain — or is a switchback just a learning rate wearing a costume?

§2 The machine

The fit skeleton is the house logistic readout verbatim (27-way softmax, batch 8192, gradient clip 1.0, EWA momentum [1] $\beta=0.9$, validation early stopping, and the AUTO-VOID divergence tripwire). The switchback step is a *unit* step tilted downhill at fixed grade:

```
#  $\hat{m}$  = normalized momentum (the fall line)
#  $\hat{u}$  = persistent traverse direction, re-leveled every step
#      (orthogonalized against  $\hat{m}$ , renormalized)
#  $\sigma \in \{+1, -1\}$  flips every L steps – a hairpin
d = grade ·  $\hat{m}$  +  $\sqrt{1 - \text{grade}^2}$  ·  $\sigma$  ·  $\hat{u}$ 
W ← W – s · d
# effective downhill stride:  $\eta_{\text{eff}} = s \cdot \text{grade}$ 
# traverse amplitude:      A = s · L · cos  $\theta$ 
```

At grade = 1 the machine reduces *bitwise* to the promoted fixed-stride rope machine of the goat study (its noprobe arm) — verified before registration: the grade-1 smoke reproduces the on-disk values to four decimals on both terrains. Straight-descent anchors at matched η_{eff} therefore serve as exact controls, and every cell is scored as $\Delta = \text{switchback val bpc} - \text{anchor val bpc}$ at the same η_{eff} , within terrain and seed.

Terrains as in Goat Descent: **champ**, the standard mixing pond's reservoir features [3] (sparse tanh, $\rho=0.8$), and **cliff**, the addressed pond's *linear* variant at $\rho=0.95$ — the most ill-conditioned features in the lab's inventory (see whitepaper 01). Standard text8 splits [4], 2M train characters. Nine cells per terrain and seed: six matched- η_{eff} (s, grade) pairs spanning amplitudes $A \approx 0.06$ to 15.9 at L=16, two leg-length cells (L=4 and L=64 at s=0.1, grade=0.1), and a fresh grade-1 anchor at s=1e-2.

§3 What we registered in advance

Five ordinal predictions and two bands at seed 0, each with named alternatives — and as in the goat study, the finding arrived partly through the misses:

- **S1** (a learning rate in disguise): ≥ 7 of the 8 small-amplitude cells ($A \leq 0.7$) tie their matched anchor within ± 0.01 bits. (Hit, 8/8.)
- **S2** (the traverse tax — the registered headline): all 6 large-amplitude cells ($A \geq 6$) pay $> +0.01$ bits. Named loss: wide traverses tie or win, making the switchback a learning rate at every amplitude. (Missed — *and the named loss's own reading was falsified too*; see §4.)
- **S3** (amplitude collapse): a stride-built and a leg-built cell of matched amplitude land within a stated band of each other — A is the knob, not stride or leg separately. (Half-falsified, decisively on champ.)
- **S4** (terrain order): the ill-conditioned cliff punishes traversing harder at every large-amplitude setting. (Missed, 0/3 — direction *reversed*.)
- **S5** (the rope holds): zero AUTO-VOIDs among the ten non-large-amplitude cells. (Hit; in fact 0/18 overall.)

The seed-1 rerun then registered the surviving structure as R1–R4 with a promotion rule stated before running: R1 (both cliff stride-built wide cells rescue, $\Delta < -0.01$), R2 (both L=64 cells taxed, $\Delta > +0.03$), and R3 (small-amplitude ties, $\geq 7/8$) all hitting would promote; any split would be journaled without promotion. R4 registered within- η_{eff} monotonicity of the tax on champ. Bands RQ1–RQ3 put numbers on the rescue, the tax, and the fresh anchors.

§4 What happened

Table 1. Seed 0 grid (run ledger #10). Δ = switchback val bpc – matched- η_{eff} straight-descent anchor. Zero AUTO-VOIDs in 18/18 cells.

cell (s, grade, L)	A	η_{eff}	champ val / Δ	cliff val / Δ
(4e-3, .25, 16)	0.062	1e-3	3.0169 / +0.0004	3.1752 / +0.0003
(1e-2, .1, 16)	0.159	1e-3	3.0191 / +0.0026	3.1769 / +0.0020
(1e-1, .1, 4)	0.40	1e-2	2.7261 / +0.0011	3.1671 / -0.0001
(4e-2, .25, 16)	0.62	1e-2	2.7273 / +0.0023	3.1671 / -0.0001
(1e-1, .1, 16) mid	1.59	1e-2	2.7354 / +0.0104	3.1688 / +0.0016
(4e-1, .25, 16)	6.20	1e-1	2.6395 / +0.0011	3.2256 / -0.0185
(1e-1, .1, 64)	6.37	1e-2	2.8131 / +0.0881	3.2480 / +0.0808
(1, .1, 16)	15.92	1e-1	2.6439 / +0.0055	3.2221 / -0.0220

The bolded rows carry the study. S1 hit 8/8 — every small-amplitude cell ties its anchor, $\max |\Delta| = 0.0026$. S2 missed: only the two L=64 cells (of six large-amplitude cells) show the predicted tax, and its named loss — "a learning rate in costume at *every* amplitude" — is contradicted by those same L=64 taxes. The registered variable A splits *by knob*: stride-built amplitude is free (or better), leg-built amplitude is taxed. S3's champ half was decisively falsified: stride-built A=6.20 and leg-built A=6.37 differ by 0.1736 bits against a 0.03 band — an $\sim 80\times$ tax ratio at matched amplitude. (Its cliff half hit the registered band but was flagged in the journal as a band-width artifact: the two cells sit on opposite sides of zero.) S4 reversed outright: at every large-amplitude setting the cliff punished traversing *less* than champ, and at both stride-built settings the cliff traverse was negative — it *helped*. That rescue — two independent cells, same direction, well outside the tie band — was the run's surprise, held provisional under house rules. Of the seed-0 bands, the fresh-anchor band hit 2/2; the amplitude-power-law band was unjudgeable by its own registered gate (only two qualifying cells, where the gate required three).

Table 2. The registered replication, seed 1 (run ledger #11): 4/4 ordinals, 3/3 bands. Fresh pond draws and batch order; 0/18 AUTO-VOIDs.

claim	seed 0	seed 1
R1 cliff rescue, stride-built wide cells (Δ)	-0.0185 · -0.0220	-0.0102 · -0.0251
R2 leg tax, L=64 (Δ champ · cliff)	+0.0881 · +0.0808	+0.0897 · +0.0756
R3 small-amplitude ties ($\max \Delta $ of 8)	0.0026	0.0022
R4 champ tax monotone within $\eta_{\text{eff}}=1e-2$	+0.0881 > +0.0104 > +0.0017*	+0.0897 > +0.0100 > +0.0015*

(*The last term is the mean of the two smallest-amplitude cells at that η_{eff} , as registered — the mean absorbs the noise floor.) One honesty flag, recorded as registered: seed 1's weaker rescue cell landed at $\Delta = -0.0102$, just 0.0002 inside the registered -0.01 line — thin, flagged in the journal, judged as written. Promotion followed the pre-stated $R1 \wedge R2 \wedge R3$ rule on two independent passes.

Two scope notes. First, the rescue is *of the operating point, not of the terrain* — it is not a terrain championship: the goat study's best cliff run (goat, 3.1647) still leads the terrain, and the rescued cells (3.2221–3.2256) sit far above both it and the fixed-stride rope at $\eta=1e-3$ (3.1749). What the wide fast traverse rescues is the big-stride operating point specifically (straight descent at $\eta_{\text{eff}}=0.1$ sits at 3.2441 / 3.2437 on the two seeds). Second, the candidate collapse variable for the leg tax — the leg aspect ratio $A/\eta_{\text{eff}} = L \cdot \cos \theta / \text{grade}$, ~ 637 in the taxed cells against 62–159 in the free wide cells — was registered only as the within- η_{eff} monotonicity above (R4, both seeds). The aspect-ratio law itself, and the decomposition of the dither, are named follow-ups, not results.

§5 What it means

A switchback is a learning rate in disguise. At matched effective downhill stride, the grade cap does nothing that the stride didn't already do — sixteen of sixteen small-amplitude cells tie their straight-descent controls. Whatever intuition says about the safety of a gentle grade, on a convex fitting problem it prices as $\eta_{\text{eff}} = s \cdot \text{grade}$ and nothing else.

The tax is dawdling, not width. Where the traverse does cost bits, the cost tracks how *long* the legs are relative to descent, not how wide the trail swings: legs of 64 steps pay +0.076–0.090 bits on both terrains and both seeds, while stride-built swings of the same amplitude are free. A plausible mechanism, recorded in the journal as unregistered color: a slow leg gives the momentum estimate time to contaminate — the restoring gradient from the traverse displacement enters the fall-line estimate, and the downhill budget is spent fighting the trail. A fast leg reverses before the contamination matures.

On the scree slope, dancing beats marching. The ill-conditioned terrain was supposed to punish traversing hardest; instead, fast wide dither *rescued* 0.010–0.025 bits of the ~ 0.07 -bit big-stride thrash penalty, in all four cells across both seeds — at the same operating point where the goat study observed divergence that never tripped the void net. Orthogonal high-frequency dither appears to act as a smoothing perturbation on oscillating dynamics — a cousin of the broader observation that deliberately injected noise during training can act as a regularizer [2]. The mountain metaphor inverted twice: width is not the danger, dawdling is; and on the crumbling face, crossing the fall line beats walking it.

§6 References

1. B. T. Polyak, "Some methods of speeding up the convergence of iteration methods," *USSR Computational Mathematics and Mathematical Physics* 4(5), 1964.
2. N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, "Dropout: a simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research* 15(56), 2014.

3. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
4. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **00764148** — 2026-08-11-caity-goat-ii: the switchback brief; 18 fits + 6-fit smoke, seed o (run ledger #10). Registered before running; the grade-1 identity gate and the smoke's void behavior disclosed in the registration.
- **9e62e253** — 2026-08-11-switchback-rerun1: full 18-fit grid on seed 1, fresh pond draws and batch order (run ledger #11). Promotion by the pre-stated $R1 \wedge R2 \wedge R3$ rule on two independent passes.

All claims judged strictly against the registered wording, misses included above; 2 seeds $\times$ 2 terrains; replicated before publication.

AuxiLab · findings are pre-registered and replicated before publication

© 2026 Caitlyn Meeks — Made in the Canary Islands 